EXP NO. 4	DESIGN AND IMPLEMENTATION OF PARITY GENERATOR (CHECKERS)	DATE: 11/10/22

AIM:

To design and impliment 3-bit parity generator and checkers.

APPARATUS REQUIRED:

S.NO	NAME OF THE COMPONENT	SPECIFICATION	QUANTITY
1	IC trainer kit	-	1
2.	Logic gates IC's	IC 7486	1
3	Connecting wires	-	1 set

THEORY:

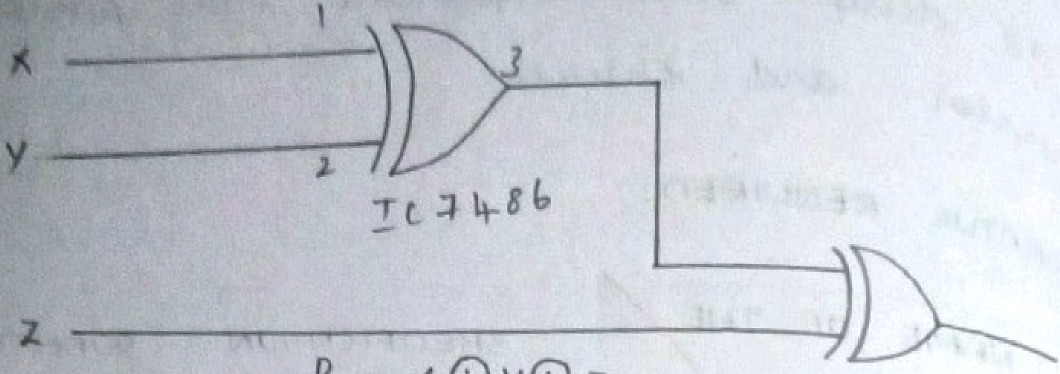
(*) A parity bit is used for the purpose of detecting error during transmission of binary information.

(*) A parity bit is an extra single bit included with a message.

(*) An error is detected if the checker parity does not correspond with the one transmitted.

(*) The parity which is transmitted along with the message is called parity

EVEN PARITY GENERATOR LOGIC DIAGRAM



$P = X \oplus Y \oplus Z$

TRUTH TABLE:

X	Y	Z	P
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

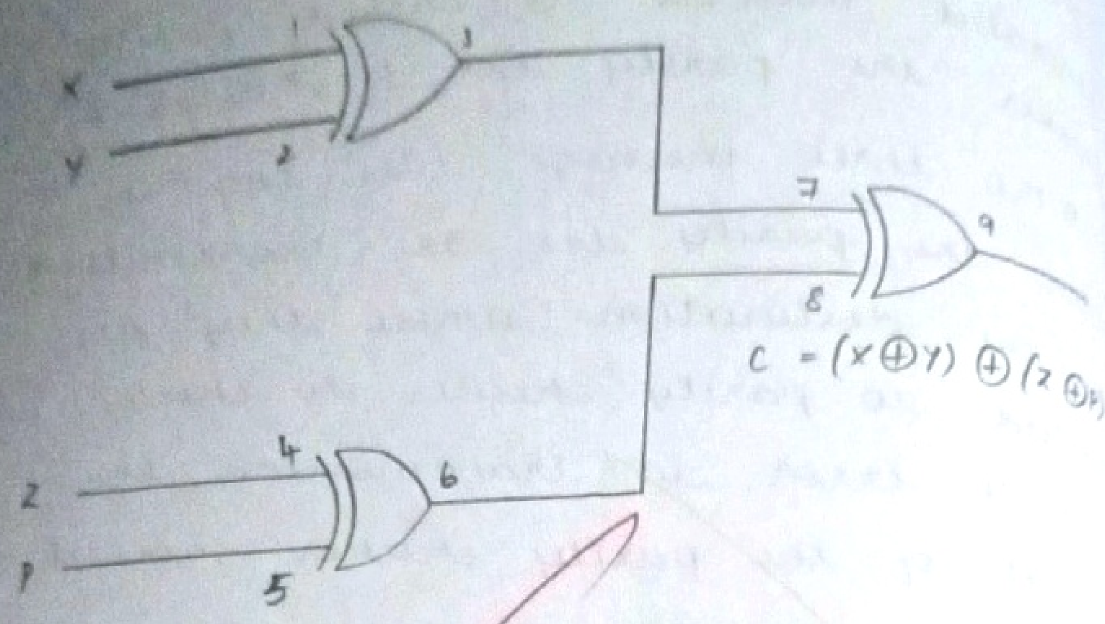
generator the parity which is generated received is called parity checker. the parity bit $P = x \oplus y \oplus z$.

(*) The three message bits together with the parity bit are transmitted to the destination where they are applied to parity checker to check possible error in transmission the output of the parity checker should have even number of ones. for accurate transmission.

(*) If the error is encounter then it will have odd number of ones the parity checker can be implemented with exclusive or OR gate.

$$C = X \oplus Y (Z \oplus P).$$

EVEN PARITY CHECKER LOGIC DIAGRAM



TRUTH TABLE:

X	Y	Z	P	C
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	01
1	0	0	1	0
1	0	1	0	0
1	0	1	1	01
1	1	0	0	0
1	1	0	1	01
1	1	1	0	01
1	1	1	1	0

PROCEDURE:

(*) Connections are made as per the circuit diagram

(*) Switch ON IC trainer kit

(*) Feed the logic 0 or 1 to the input variable.

(*) The output is verified using the LED indicators.

RESULT:

Thus 3 bit parity generator and checkers were designed and impemented.

EXP NO: 5	DESIGN AND IMPLEMENTATION OF PRIORITY ENCODERS USING LOGIC GATES	DATE: 11/10/22

AIM:

To design and implement priority encoder using logic gates.

COMPONENTS REQUIRED:

S.NO	NAME OF THE COMPONENT	SPECIFICATION	QUANTITY
1	IC Trainer kit	-	1
2	Logic gates IC's	IC 7404, IC 7432	1
3	Logic gates IC's	IC 7408	2
4	Connecting wires	-	1 set

THEORY:

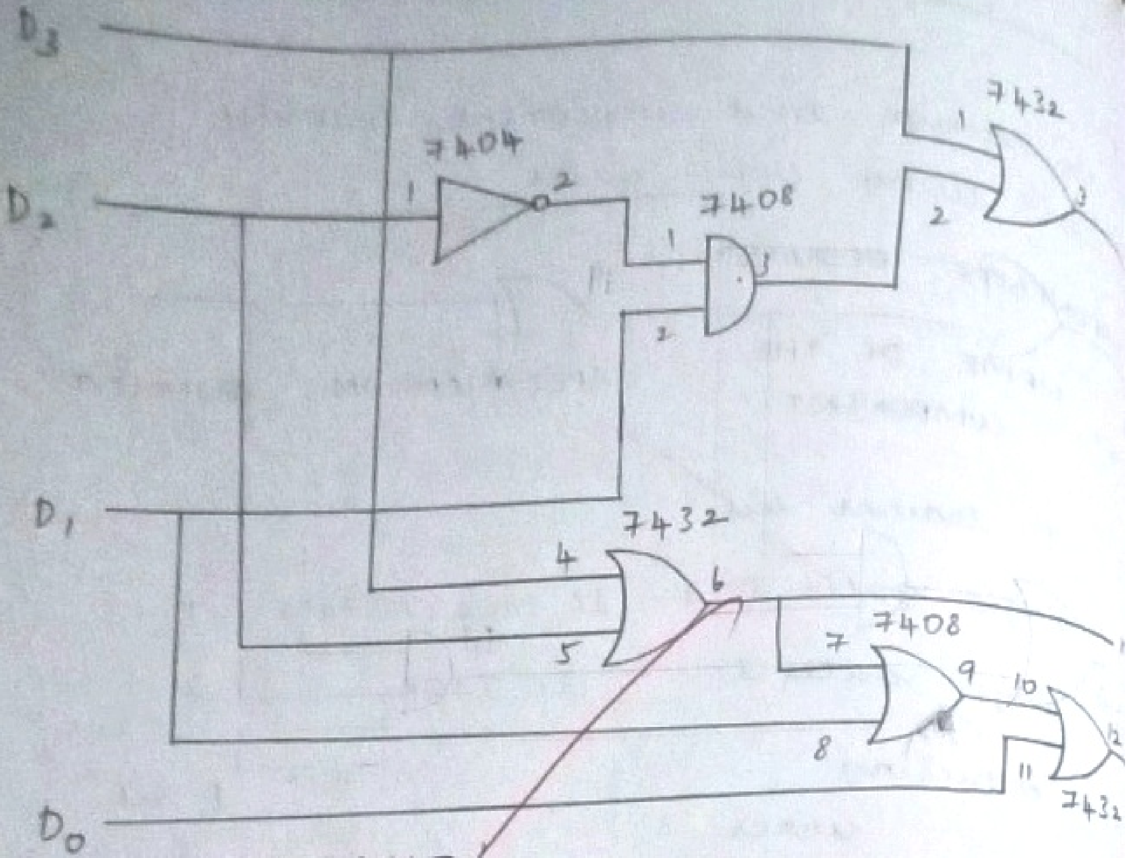
(*) An encoder is digital circuit that performs the inverse operations of the decoder.

(*) It has 2n input and n output that represent the code of the order of the input that is set to one.

(*) This encoder has two main draw backs.

(*) When more than one input is not same time then output could indicate a wrong code.

LOGIC DIAGRAM:



TRUTH TABLE:

Input				Output		
D0	D1	D2	D3	X	Y	Z
0	0	0	0	X	X	0
1	0	0	0	0	0	1
X	1	0	0	0	1	1
X	X	1	0	1	0	1
X	X	X	1	1	1	1

eg: D_5 and D_6 are one at the same time then and are once indicating D_7 is one

(*) If no input is one which is not valid code, then are all zero which indicate a code do to overcome these priority encoder

PROCEDURE:

(*) Connections are made as per the circuit diagram

(*) Switch ON the IC trainer kit

(*) Apply the logic input 0 or 1 to the binary input of priority encoder.

(*) Verify the truth table by observing the output indicators.

RESULT:

Thus the 4 to 2 priority encoder was designed and implemented using logic gates and their operations are verified.